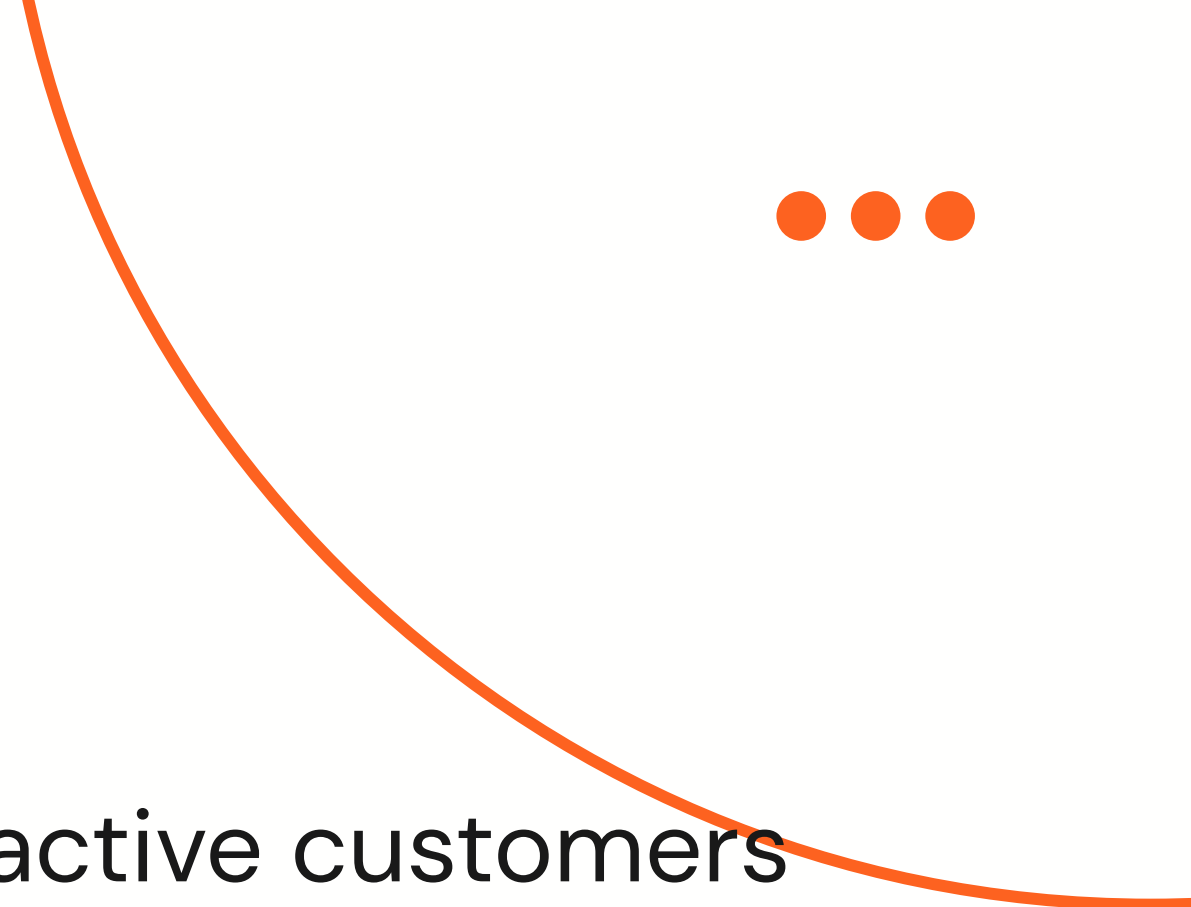


CUSTOMER CHURN ANALYSIS - TELECOM

727723EUCS132
MURALISAN J



PROBLEM STATEMENT



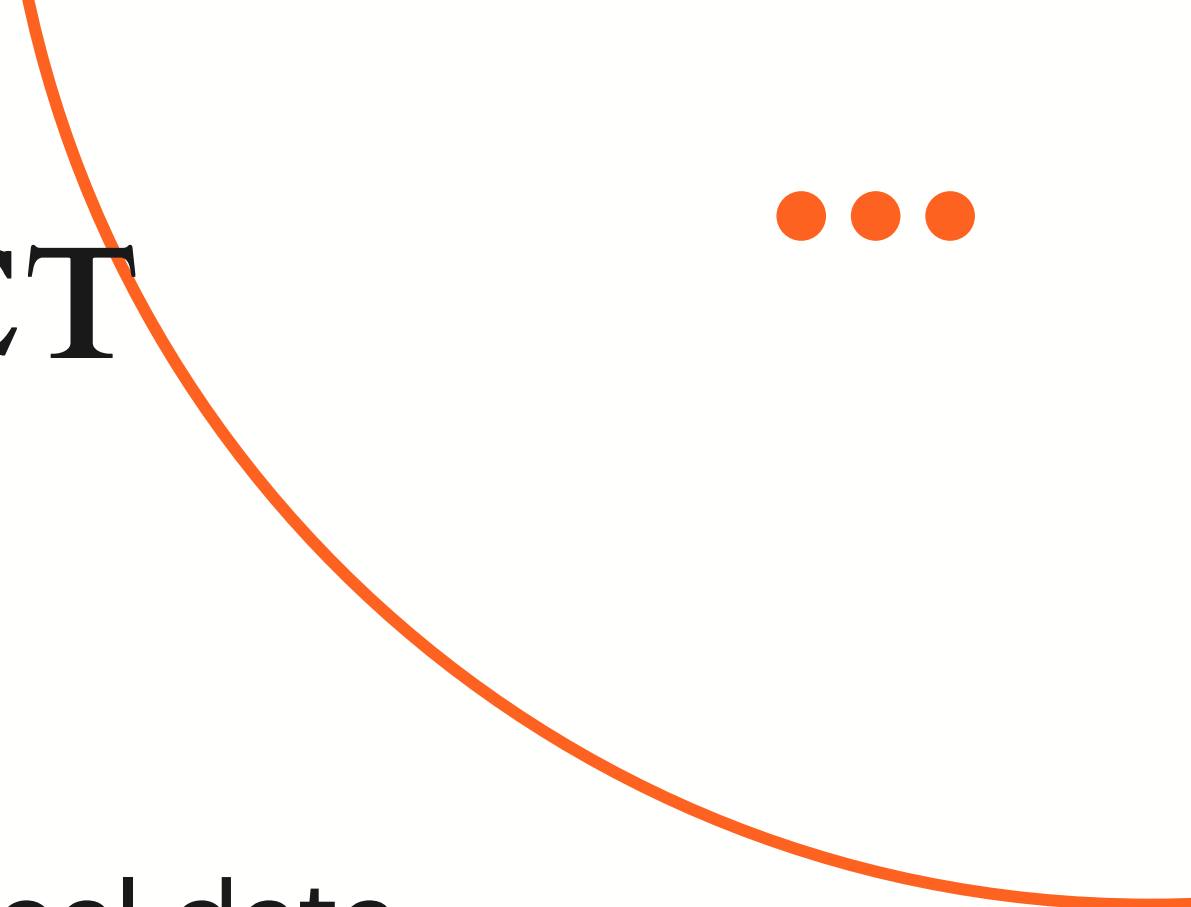
Business Problem

- A telecom company observed a decline in active customers
- Increasing customer churn is impacting monthly revenue
- Management lacks clear insights into why customers are leaving

Business Need

- Identify churn patterns
- Understand key churn drivers
- Enable proactive customer retention strategies

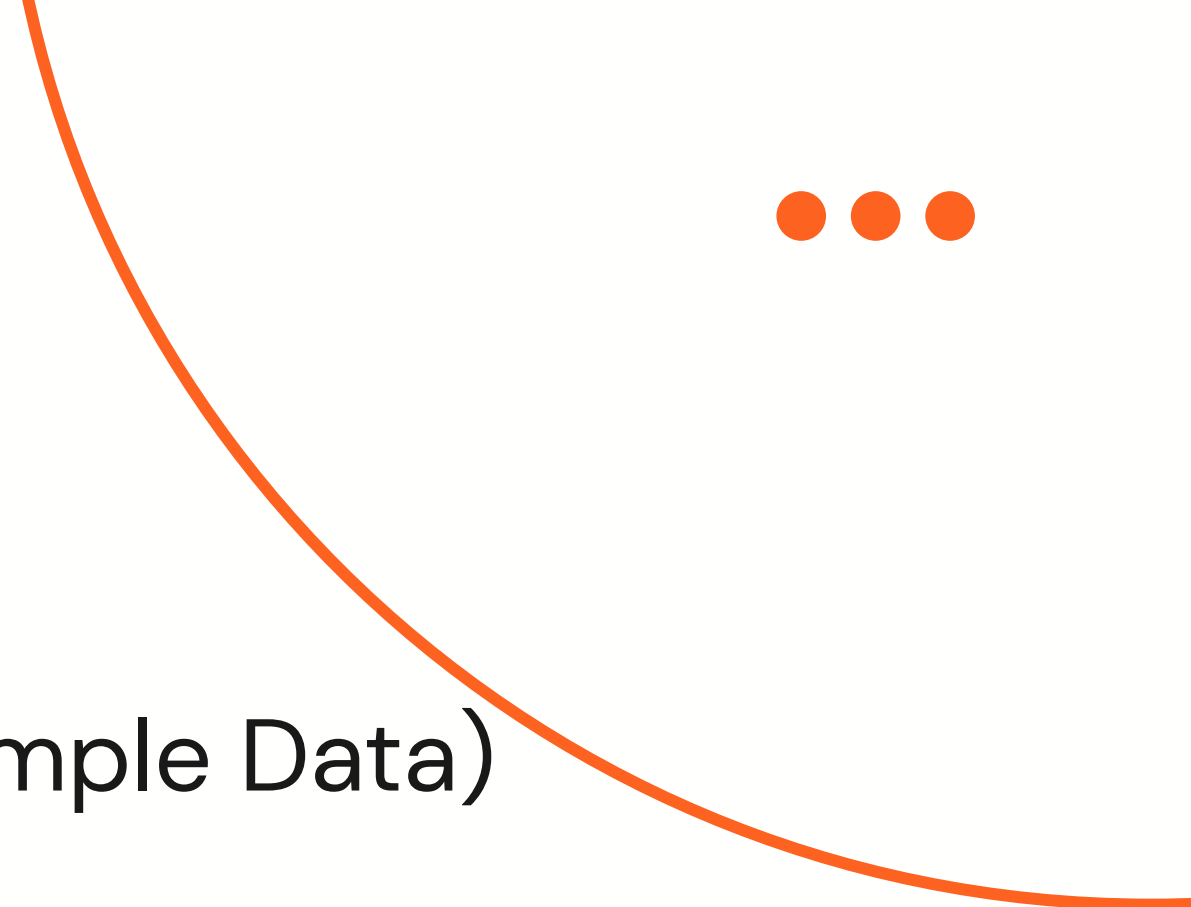
OBJECTIVES OF THE PROJECT



Project Objectives

- Analyze customer behavior using historical data
- Identify customers at high risk of churn
- Compare churn patterns for prepaid and postpaid users
- Build an interactive Power BI dashboard
- Provide actionable insights for business decisions

DATASET OVERVIEW



Dataset Used

- Telecom Customer Churn Dataset (IBM Sample Data)

Key Attributes

- Customer ID
- Gender
- Tenure (months)
- Contract type
- Monthly charges
- Billing type (Prepaid / Postpaid)
- Churn status

Customer Churn Analysis - Telecom

3157

Total Customers

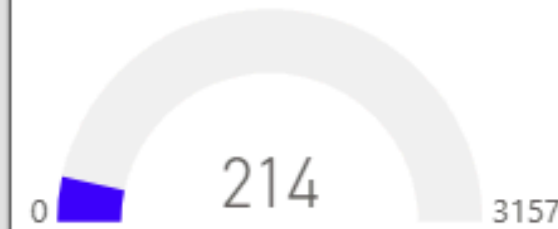
214

Churned Customers

6.78

Churn Rate (%)

Churned Customers and Total Customers



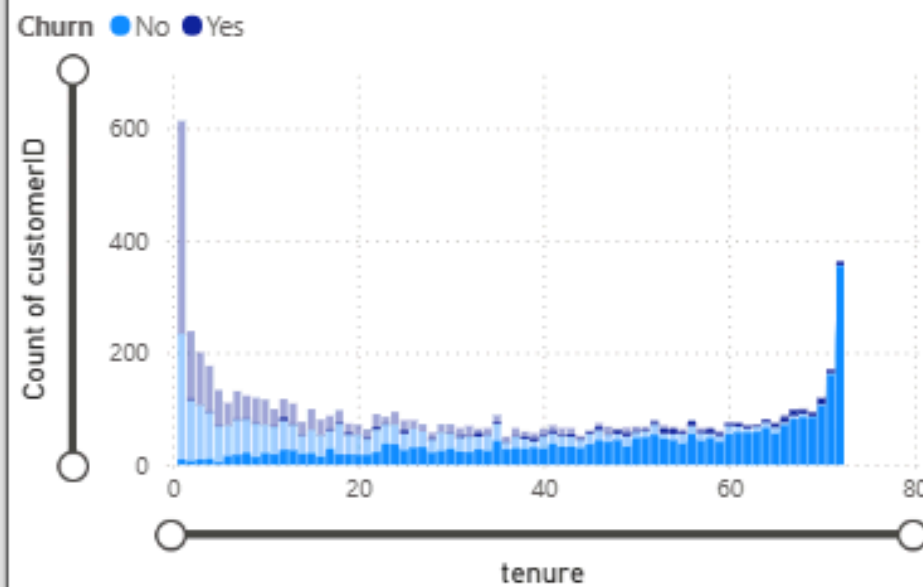
gender
☐ Female
☐ Male

Billing ...
☐ Postpaid
☐ Prepaid

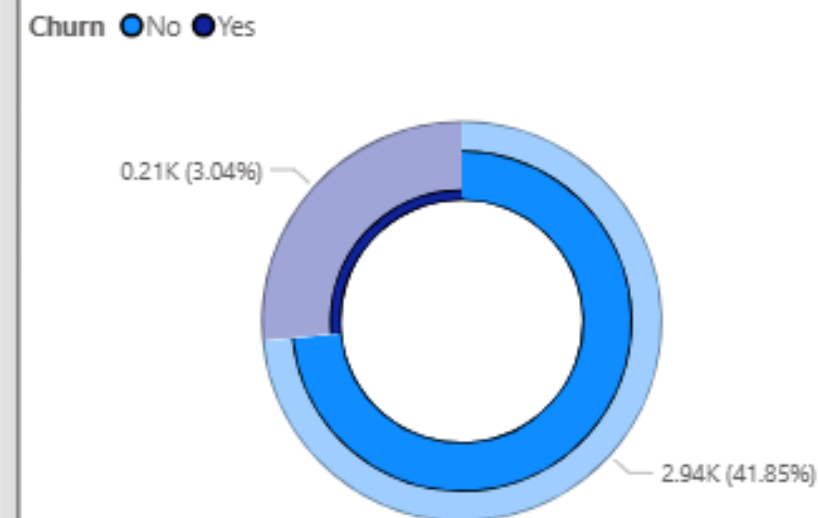
Contract
☐ Month-to...
☐ One year
☐ Two year



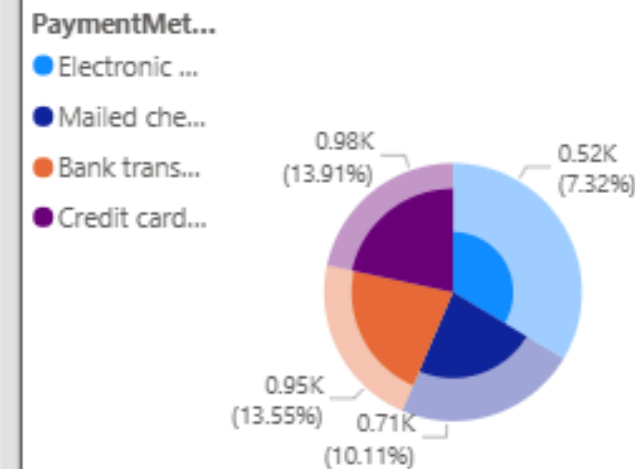
Count of customerID by tenure and Churn



Count of customerID by Churn



Count of Churn by PaymentMethod

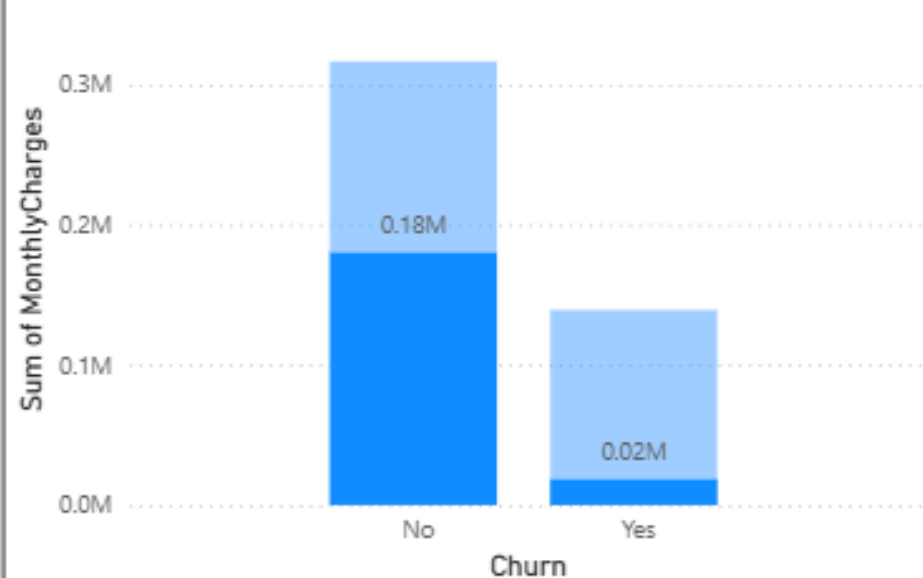


What is the churn rate by gender

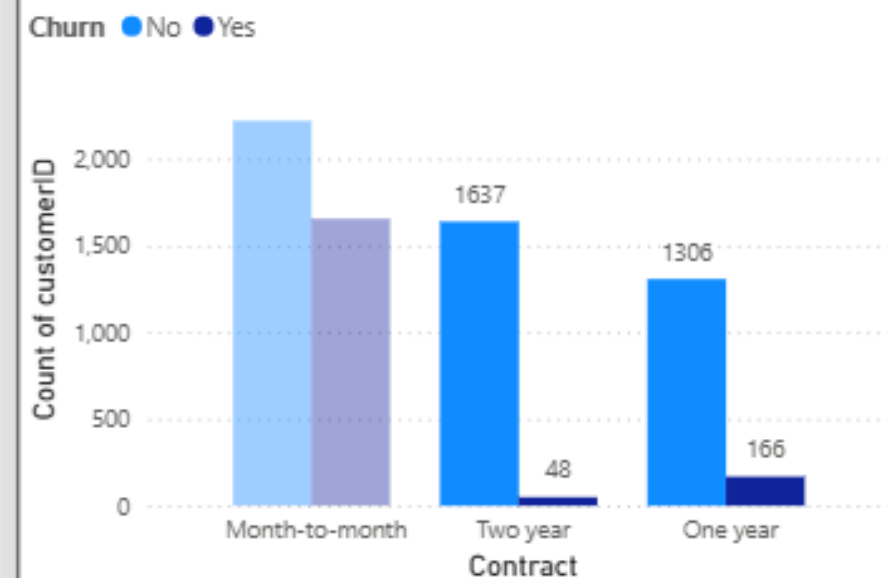


Content created by AI may be inaccurate. [Read terms](#)

Sum of MonthlyCharges by Churn



Count of customerID by Contract and Churn



Which billing type has highest churn

Showing Billing.type and top churn of WA results fn use C telco customer churns for by total total charge



Content created by AI may be inaccurate. [Read terms](#)

Total Count of customerID was higher for No (2,943) than Yes (214).

72 in Churn No made up 11.21% of Count of customerID.

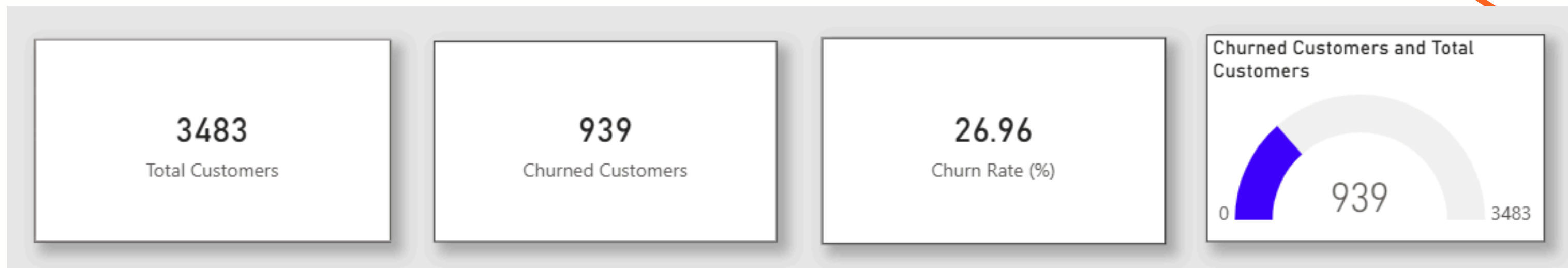
Average Count of customerID was higher for No (40.87) than Yes (3.51).

Count of customerID for No and Yes diverged the most when the tenure was 72, when No were 348 higher than Yes.

Sum of MonthlyCharges for No (1,80,083.10) was higher than Yes (18,283.75).

No accounted for 90.78% of Sum of MonthlyCharges.

KEY METRICS & DAX CALCULATIONS



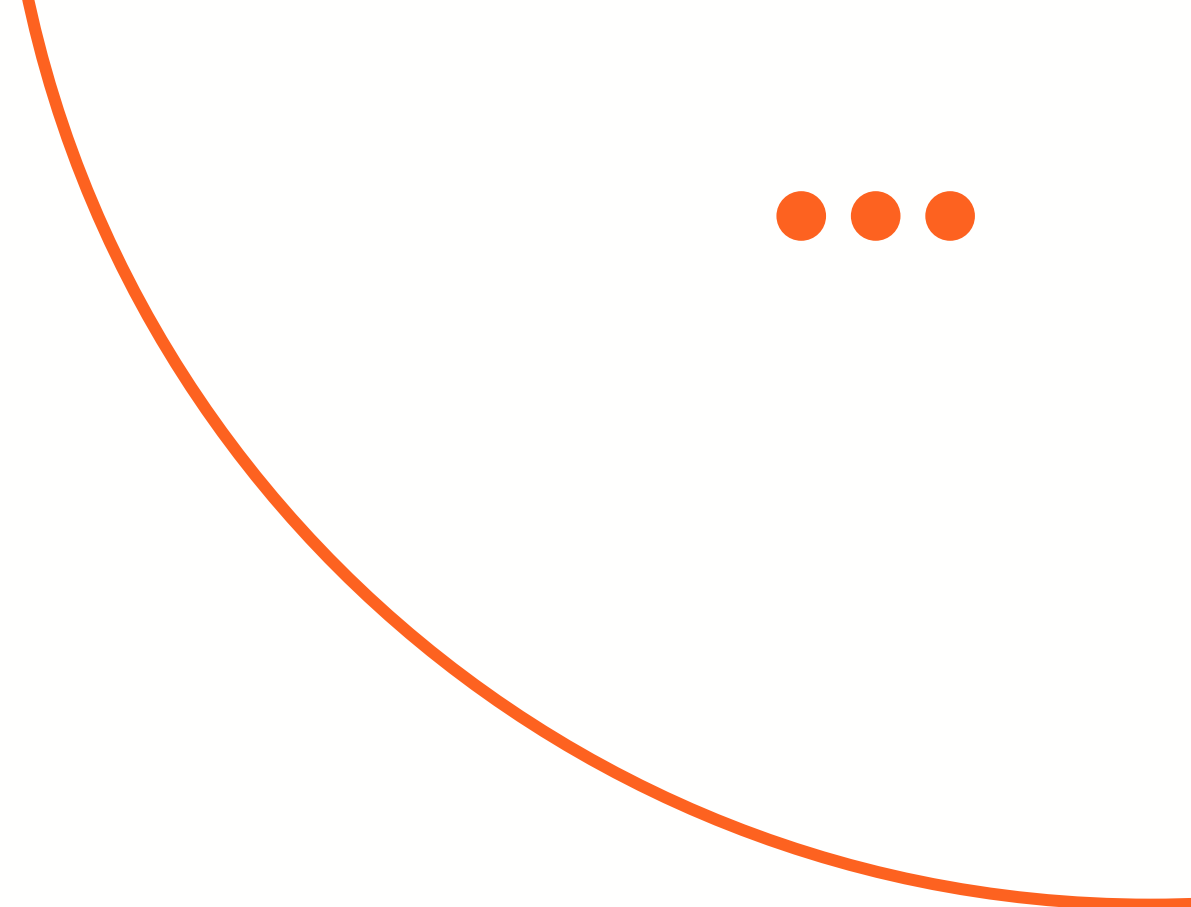
Churn Rate Formula

- **Churn Rate = (Churned Customers / Total Customers) × 100**

Overall Results

- Total Customers: 7,032
- Churned Customers: 1,869
- Churn Rate: 27%

TENURE & CONTRACT ANALYSIS

A decorative orange line starts from the top right corner, curves downwards and to the left, and then continues horizontally to the right edge of the slide. Three orange dots are positioned above the line, near the top right corner.

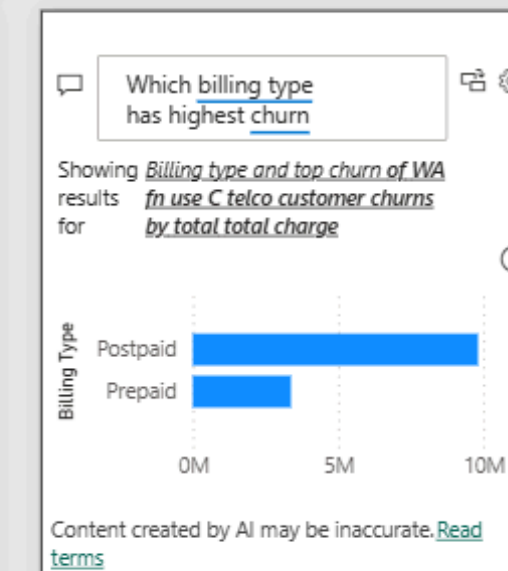
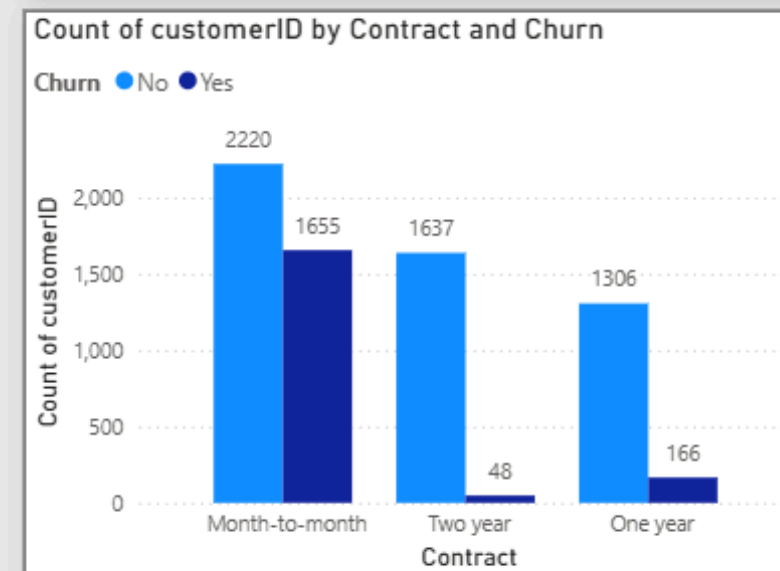
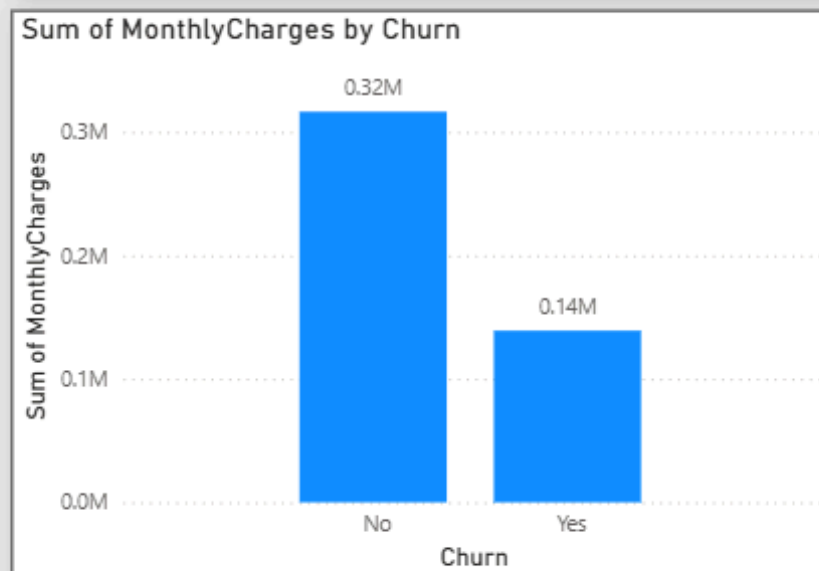
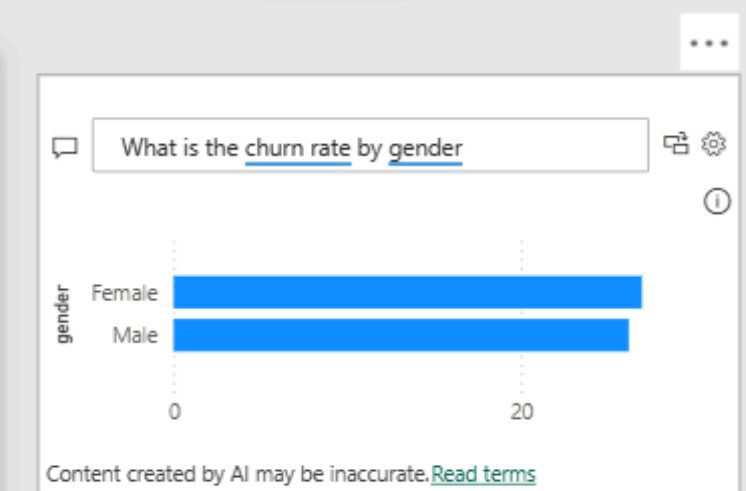
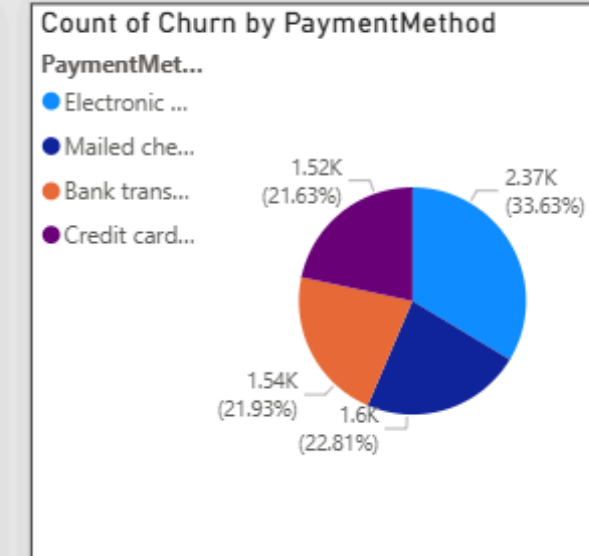
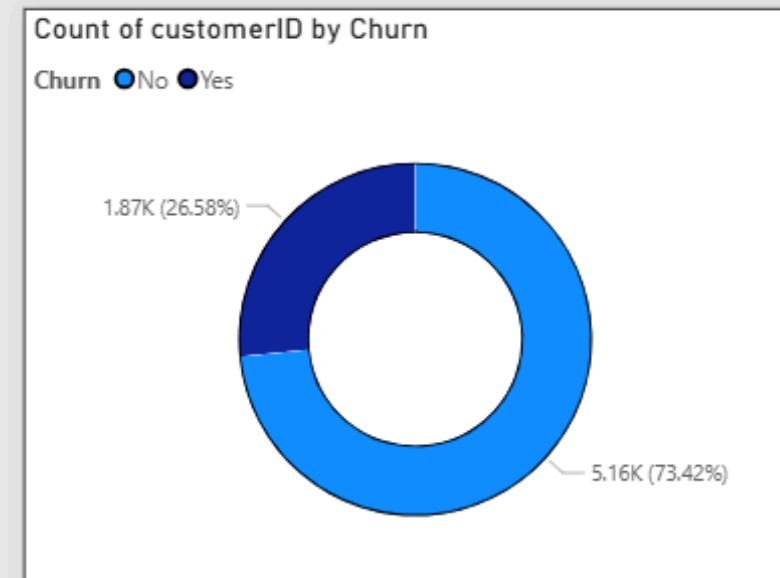
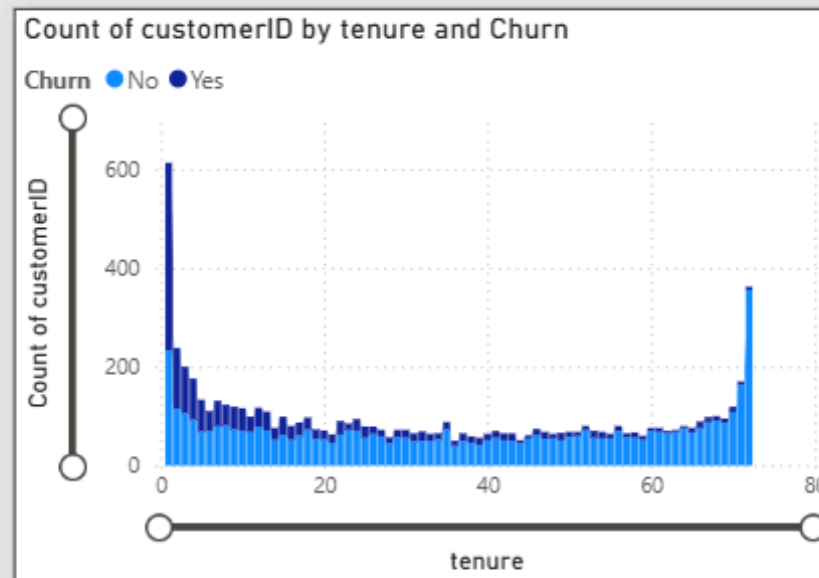
Tenure Insights

- Highest churn observed in the first 6–12 months
- Long-term customers show higher retention

Contract Type Insights

- Month-to-month contracts have the highest churn
- Two-year contracts have the lowest churn

TENURE & CONTRACT ANALYSIS



Total Count of customerID was higher for No (5,163) than Yes (1,869).

1 in Churn Yes made up 5.40% of Count of customerID.

Average Count of customerID was higher for No (71.71) than Yes (25.96).

Count of customerID for No and Yes diverged the most when the tenure was 72, when No were 350 higher than Yes.

Sum of MonthlyCharges for No (3,16,530.15) was higher than Yes (1,39,130.85).

No accounted for 69.47% of Sum of MonthlyCharges.

INTERACTIVE SLICERS & FILTERS

Slicers Used in the Dashboard

The image shows three interactive slicers arranged horizontally on a light gray background. Each slicer is a white rounded rectangle with a thin black border and a subtle drop shadow. The first slicer, labeled 'gender', contains two radio button options: 'Female' and 'Male'. The second slicer, labeled 'Billing ...', contains two radio button options: 'Postpaid' and 'Prepaid'. The third slicer, labeled 'Contract', contains three radio button options: 'Month-to...', 'One year', and 'Two year'. Each slicer has a small downward-pointing chevron icon next to its title. The 'Billing ...' slicer has a blue border, while the others have black borders.

gender

☐ Female

☐ Male

Billing ...

☐ Postpaid

☐ Prepaid

Contract

☐ Month-to...

☐ One year

☐ Two year

PREPAID VS POSTPAID ANALYSIS

Billing Type Classification

- Month-to-month → Prepaid
- One-year / Two-year → Postpaid

Key Findings

- Prepaid customers have a higher churn rate
- Postpaid customers are more stable due to long-term commitment

Reason

- Prepaid users have low commitment and high price sensitivity

CHURN DRIVERS & HIGH-RISK SEGMENT

Key Churn Drivers Identified

- Short customer tenure
- Month-to-month contracts
- High monthly charges
- Prepaid billing type

High-Risk Customer Segment

- Prepaid customers
- Tenure less than 6 months
- Higher monthly charges

CONCLUSION

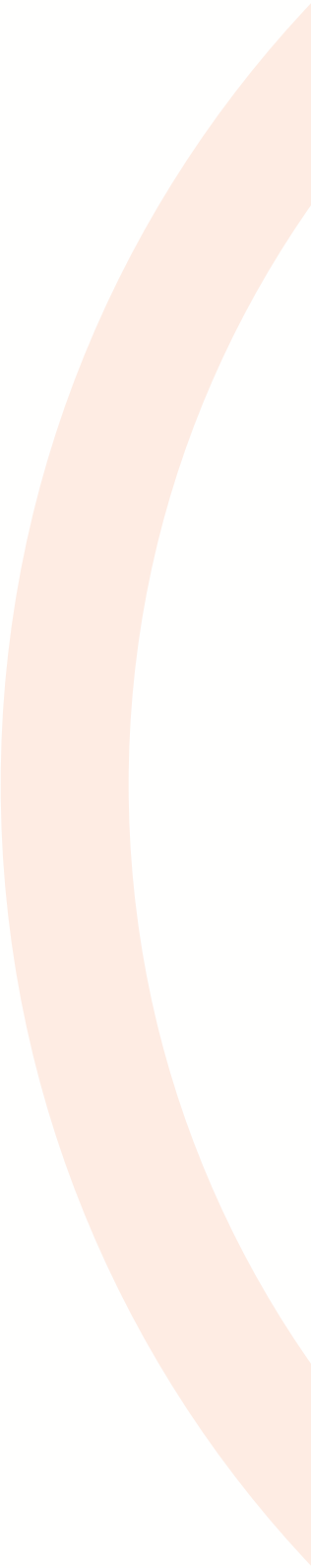
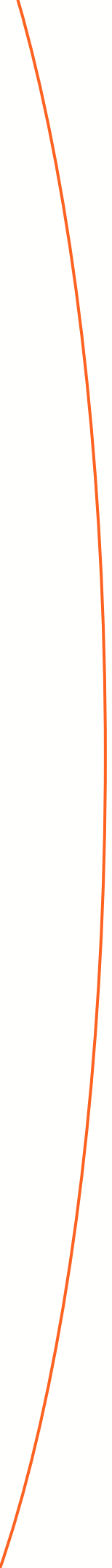


Project Summary

- The dashboard provides a clear view of churn behavior
- Identifies high-risk customers early
- Helps management make data-driven retention decisions

Final Outcome

- Reduced churn risk
- Improved customer retention
- Better revenue stability



***Thank
you !!!***